

Constraint Comparison: Original vs FormatFuzzer Templates

This document compares constraints extracted by LLM from the original binary templates (*-orig) versus the FormatFuzzer-edited templates for each file format. Each table uses a Venn-diagram layout with three columns:

- **Original Template Only:** constraints present only in the original (upstream) template
- **In Both:** constraints present in both templates (with differences noted where applicable)
- **FormatFuzzer Template Only:** constraints present only in the FormatFuzzer-edited template

Constraints shared by both templates that have different values are annotated with [DIFF] and show both values side by side.

AVI

Original constraints: 17 FormatFuzzer constraints: 22 Only in orig: 6 In both (same): 7 In both (diff): 4 Only in FF: 11

Original Template Only	In Both	FormatFuzzer Template Only
BITMAPINFOHEADER.biPlanes Fixed: 0x0001 [risk: high]	JUNKHEADER.id Fixed: 0x4A554E4B [risk: high]	AVIINDEXENTRY.dwFlags Fixed: 0x00 [risk: medium]
MainAVIHeader.dwReserved1 Fixed: 0x00000000 [risk: low]	LISTHEADER.id Fixed: 0x4C495354 [risk: high]	BITMAPINFOHEADER.biBitCount Fixed: 0x08 [risk: high]
ROOT.dataalen Length of RIFF chunk data in bytes - should equal file size minus 8 [risk: medium]	ROOT.form Fixed: 0x41564920 [risk: high]	BITMAPINFOHEADER.biCompression Fixed: 0x01 [risk: high]
WAVEFORMATEX.wFormatTag Enum: {1, 2, 3, 6, 7, 17, 85} [risk: high]	avihHEADER.id Fixed: 0x61766968 [risk: high]	MOVICHUNK.id Enum: {00db, 00dc, 00pc, 00wb} [risk: high]
avihHEADER.dataalen Fixed: 0x00000038 [risk: high]	idx1HEADER.id Fixed: 0x69647831 [risk: high]	ROOT.root_dataalen RIFF chunk size - total file size minus 8 [risk: high]
idx1HEADER.dataalen Index data length - must be a multiple of AVIINDEXENTRYLEN (16) [risk: high]	strfHEADER.id Fixed: 0x73747266 [risk: high]	VideoPropHeader.id Fixed: 0x76707270 [risk: high]
	strhHEADER.id Fixed: 0x73747268 [risk: high]	VideoPropHeader.nbFieldPerFrame Enum: {1, 2} [risk: medium]
	AVIStreamHeader.fccType [DIFF] Orig: Enum: {vids, auds, txts, mids} [risk: high] FF: Enum: {vids, auds} [risk: high]	avihHEADER.avi_hdr_dataalen Fixed: 0x38 [risk: high]
	LISTHEADER.type [DIFF] Orig: Enum: {hdrl, strl, movi, INFO, odml} [risk: high] FF: Enum: {hdrl, strl, movi} [risk: high]	strfHEADER_BIH.strf_hdr_bih_dataalen Range: [40, 58] [risk: medium]
	ROOT.id [DIFF] Orig: Enum: {RIFF, RIFX} [risk: high] FF: Fixed: 0x52494646 [risk: high]	strfHEADER_WAVE.strf_hdr_wave_dataalen Range: [20, 36] [risk: medium]
	strnHEADER.id [DIFF] Orig: Fixed: 0x7374726E [risk: medium] FF: Fixed: 0x76656474 [risk: medium]	strhHEADER.strh_hdr_dataalen Fixed: 0x38 [risk: high]

BMP

Original constraints: 13 FormatFuzzer constraints: 12 Only in orig: 1 In both (same): 1 In both (diff): 11 Only in FF: 0

Original Template Only	In Both	FormatFuzzer Template Only
BITMAPINFOHEADER.biClrUsed	BITMAPFILEHEADER.bfType	

Range: [0, 256] [risk: medium]	Fixed: 0x424D [risk: high]	
	BITMAPFILEHEADER.bfOffBits [DIFF] Orig: Offset from start of file to pixel data [risk: high] FF: Byte offset from file start to pixel data [risk: high]	
	BITMAPFILEHEADER.bfReserved1 [DIFF] Orig: Fixed: 0x0000 [risk: low] FF: Fixed: 0x00 [risk: medium]	
	BITMAPFILEHEADER.bfReserved2 [DIFF] Orig: Fixed: 0x0000 [risk: low] FF: Fixed: 0x00 [risk: medium]	
	BITMAPFILEHEADER.bfSize [DIFF] Orig: Total bitmap file size in bytes [risk: medium] FF: Total file size in bytes - must match actual file size [risk: high]	
	BITMAPINFOHEADER.biBitCount [DIFF] Orig: Enum: {1, 2, 4, 8, 16, 24, 32} [risk: high] FF: Enum: {1, 4, 8, 16, 24, 32} [risk: high]	
	BITMAPINFOHEADER.biCompression [DIFF] Orig: Enum: {0, 1, 2, 3, 4, 5} [risk: high] FF: Enum: {0=BI_RGB, 1=BI_RLE8, 2=BI_RLE4, 3=BI_BITFIELDS} [risk: high]	
	BITMAPINFOHEADER.biHeight [DIFF] Orig: Range: [-2147483648, 2147483647] [risk: high] FF: Range: [1, 65535] [risk: medium]	
	BITMAPINFOHEADER.biPlanes [DIFF] Orig: Fixed: 0x0001 [risk: high] FF: Fixed: 0x01 [risk: high]	
	BITMAPINFOHEADER.biSize [DIFF] Orig: Enum: {12, 40, 52, 56, 64, 108, 124} [risk: high] FF: Fixed: 0x28 [risk: high]	
	BITMAPINFOHEADER.biSizeImage [DIFF] Orig: Size of image data - may be 0 for BI_RGB but must match pixel data length when c [risk: medium] FF: Size of raw bitmap data in bytes [risk: medium]	
	BITMAPINFOHEADER.biWidth [DIFF] Orig: Range: [1, 2147483647] [risk: high] FF: Range: [1, 65535] [risk: medium]	

GIF

Original constraints: 18 FormatFuzzer constraints: 20 Only in orig: 2 In both (same): 16 In both (diff): 0 Only in FF: 4

Original Template Only	In Both	FormatFuzzer Template Only
DATASUBBLOCK.Size Sub-block size in bytes - must match the following data byte count (1..255) [risk: high]	APPLICATIONEXTENTION.ApplicationLabel Fixed: 0xFF [risk: high]	IMAGEDATA.LZWMinimumCodeSize Enum: {8, 11} [risk: high]
IMAGEDESCRIPTOR_PACKEDFIELDS.Reserved Fixed: 0x00 [risk: medium]	APPLICATIONEXTENTION.ExtensionIntroducer Fixed: 0x21 [risk: high]	IMAGEDESCRIPTOR.PackedFields.Reserved Fixed: 0x00 [risk: low]
	APPLICATIONSUBBLOCK.BlockSize	LogicalScreenDescriptor.PackedFields.Reserved

	Fixed: 0x0B [risk: high]	rved Fixed: 0x00 [risk: low]
	COMMENTEXTENSION.CommentLabel Fixed: 0xFE [risk: high]	UNDEFINEDDATA.ExtensionIntroducer Fixed: 0x21 [risk: high]
	COMMENTEXTENSION.ExtensionIntroducer Fixed: 0x21 [risk: high]	
	DATASUBBLOCKS.BlockTerminator Fixed: 0x00 [risk: high]	
	GIFHEADER.Signature Fixed: 0x474946 [risk: high]	
	GIFHEADER.Version Enum: {87a, 89a} [risk: high]	
	GRAPHICCONTROLEXTENSION.BlockTerminator Fixed: 0x00 [risk: high]	
	GRAPHICCONTROLEXTENSION.ExtensionIntroducer Fixed: 0x21 [risk: high]	
	GRAPHICCONTROLEXTENSION.GraphicControlLabel Fixed: 0xF9 [risk: high]	
	GRAPHICCONTROLSUBBLOCK.BlockSize Fixed: 0x04 [risk: high]	
	IMAGEDESCRIPTOR.ImageSeperator Fixed: 0x2C [risk: high]	
	PLAINTEXTTEXTENTION.ExtensionIntroducer Fixed: 0x21 [risk: high]	
	PLAINTEXTTEXTENTION.PlainTextLabel Fixed: 0x01 [risk: high]	
	TRAILER.GIFTrailer Fixed: 0x3B [risk: high]	

JPG

Original constraints: 21 FormatFuzzer constraints: 24 Only in orig: 11 In both (same): 4 In both (diff): 6 Only in FF: 14

Original Template Only	In Both	FormatFuzzer Template Only
APP1.EXIF.byteOrder Enum: {II, MM} [risk: high]	APP0.marker Fixed: 0xFFE0 [risk: high]	APP0.szSection Fixed: 0x10 [risk: high]
APP1.identifier Enum: {Exif} [risk: high]	DHT.marker Fixed: 0xFFC4 [risk: high]	APP0.versionHigh Fixed: 0x01 [risk: medium]
COMMENT.marker Fixed: 0xFFFE [risk: medium]	DQT.marker Fixed: 0xFFDB [risk: high]	APP0.versionLow Fixed: 0x01 [risk: medium]
DRI.marker Fixed: 0xFFDD [risk: medium]	SOS.marker Fixed: 0xFFDA [risk: high]	APP0.xThumbnail Fixed: 0x00 [risk: medium]
DRI.szSection Fixed: 0x0004 [risk: high]	APP0.identifier [DIFF] Orig: Enum: {JFIF, JFXX} [risk: high] FF: Fixed: 0x4A46494600 [risk: high]	APP0.yThumbnail Fixed: 0x00 [risk: medium]
JPGFILE.EOIMarker Fixed: 0xFFD9 [risk: high]	APP0.units [DIFF] Orig: Enum: {0, 1, 2} [risk: medium] FF: Fixed: 0x01 [risk: medium]	APP2.marker Fixed: 0xFFE2 [risk: high]
JPGFILE.SOIMarker Fixed: 0xFFD8 [risk: high]	APP1.EXIF.tagMark [DIFF] Orig: Fixed: 0x002A [risk: high] FF: Fixed: 0x2A [risk: high]	COMPSOF.HorzVert Fixed: 0x11 [risk: medium]

M_ID JPEG section markers all start with byte 0xFF followed by a non-zero marker byte [risk: high]	APP1.marker [DIFF] Orig: Fixed: 0xFFE1 [risk: high] FF: Fixed: 0xFFE0 [risk: high]	JPGFile.StartMarker Fixed: 0xFFD8 [risk: high]
SOFx.marker Enum: {65472=M_SOF0, 65473=M_SOF1, 65474=M_SOF2, 65475=M_SOF3} [risk: high]	SOS.nr_comp [DIFF] Orig: Range: [1, 4] [risk: high] FF: Enum: {1=Grayscale, 3=YCbCr} [risk: high]	SOF.marker Fixed: 0xFFC0 [risk: high]
SOFx.nr_comp Range: [1, 4] [risk: high]	szSection [DIFF] Orig: Segment length (big-endian uint16) - must equal segment content size including t [risk: high] FF: Segment length - must match actual segment content [risk: high]	SOF.nr_comp Enum: {1=Grayscale, 3=YCbCr} [risk: high]
SOFx.precision Enum: {8, 12, 16} [risk: high]		SOF.precision Fixed: 0x08 [risk: high]
		SOS.AhAI Fixed: 0x00 [risk: medium]
		SOS.Se Fixed: 0x3F [risk: medium]
		SOS.Ss Fixed: 0x00 [risk: medium]

MIDI

Original constraints: 10 FormatFuzzer constraints: 8 Only in orig: 5 In both (same): 3 In both (diff): 2 Only in FF: 3

Original Template Only	In Both	FormatFuzzer Template Only
DeltaTime.byte Variable-length delta-time byte (low 7 bits are value, high bit is continuation; [risk: high])	MidiHeader.m_format Enum: {0=MIDI_SINGLE, 1=MIDI_MULTIPLE, 2=MIDI_PATTERN} [risk: high]	DeltaTime.t0 Range: [0, 127] [risk: medium]
MidiHeader.m_seclen Fixed: 0x00000006 [risk: high]	MidiHeader.m_magic Fixed: 0x4D546864 [risk: high]	MidiHeader.m_seclen_hdr Fixed: 0x06 [risk: high]
MidiHeader.m_tickdiv Range: [1, 32767] [risk: medium]	MidiTrack.m_magic Fixed: 0x4D54726B [risk: high]	MidiTrack.m_seclen_trk Track section length - must match track data length [risk: high]
MidiMessage.meta_event.m_type Enum: {0=META_SEQUENCE_NUM, 1=META_TEXT, 2=META_COPYRIGHT, 3=META_SEQUENCE_NAME, 4=META_INSTRUMENT_NAME (+13 more)} [risk: high]	MidiHeader.m_ntracks [DIFF] Orig: Number of MTrk chunks following the header - must match actual track count [risk: high] FF: Range: [0, 8] [risk: medium]	
MidiTrack.m_seclen MTrk chunk length in bytes - must equal byte size of contained events [risk: high]	MidiMessage.m_status [DIFF] Orig: MIDI status byte - high bit (0x80) must be set when present; otherwise running s [risk: high] FF: MIDI status byte - high bit must be set when present [risk: high]	

MP3

Original constraints: 15 FormatFuzzer constraints: 19 Only in orig: 11 In both (same): 2 In both (diff): 2 Only in FF: 15

Original Template Only	In Both	FormatFuzzer Template Only
ID3v2_4_FRAME.encoding Enum: {0=ISO_8859_1, 1=UTF_16_with_BOM, 2=UTF_16BE_without_BOM, 3=UTF_8} [risk: medium]	ID3v1_TAG.id Fixed: 0x544147 [risk: high]	ID3v1_TAG.zero Fixed: 0x00 [risk: medium]

ID3v2_FRAME.id ID3v2 frame identifier - 4 uppercase ASCII alphanumeric characters [risk: high]	ID3v2_HEADER.head Fixed: 0x494433 [risk: high]	ID3v2_HEADER.i_experiment_tag Fixed: 0x00 [risk: low]
MPEG_HEADER.bitrate_index Range: [1, 14] [risk: high]	ID3v2_HEADER.ver_major [DIFF] Orig: Enum: {2, 3, 4} [risk: high] FF: Range: [3, 4] [risk: high]	ID3v2_HEADER.i_extend_head_pres Fixed: 0x00 [risk: medium]
MPEG_HEADER.channel_mode Enum: {0=Stereo, 1=JointStereo, 2=DualChannel, 3=Mono} [risk: medium]	ID3v2_HEADER.ver_revision [DIFF] Orig: Range: [0, 254] [risk: high] FF: Range: [2, 4] [risk: medium]	ID3v2_HEADER.i_unsyn_used Fixed: 0x00 [risk: medium]
MPEG_HEADER.emphasis Enum: {0, 1, 3} [risk: low]		ID3v2_HEADER.reserv_flags Fixed: 0x00 [risk: medium]
MPEG_HEADER.frame_sync Fixed: 0xFFE [risk: high]		ID3v2_HEADER.size Range: [0, 268435455] [risk: high]
MPEG_HEADER.frequency_index Enum: {0, 1, 2} [risk: high]		MPEG_FRAME.crc16 Optional CRC-16 immediately following header (when protection_bit=0) [risk: high]
MPEG_HEADER.layer_id Enum: {1, 2, 3} [risk: high]		MPEG_HDR.bitrate_index Range: [1, 14] [risk: high]
MPEG_HEADER.mpeg_id Enum: {1=MPEG-2, 2=MPEG-1} [risk: high]		MPEG_HDR.emphasis Enum: {0, 1, 3} [risk: low]
MPEG_HEADER.protection_bit Enum: {0, 1} [risk: medium]		MPEG_HDR.frame_sync Fixed: 0x0FFF [risk: high]
synchsafe_integer.raw ID3v2 synchsafe integer - each byte must have MSB cleared (< 0x80) [risk: high]		MPEG_HDR.full_hdr Composed 32-bit frame header - must match assembled subfields [risk: high]
		MPEG_HDR.layer_id Enum: {1=Layer3, 2=Layer2, 3=Layer1} [risk: high]
		MPEG_HDR.mpeg_version_id Enum: {0=MPEG_v2, 1=MPEG_v1} [risk: high]
		MPEG_HDR.protection_bit Enum: {0=protected, 1=unprotected} [risk: medium]
		MPEG_HDR.sampling_freq_index Range: [0, 2] [risk: high]

MP4

Original constraints: 18 FormatFuzzer constraints: 35 Only in orig: 12 In both (same): 0 In both (diff): 6 Only in FF: 29

Original Template Only	In Both	FormatFuzzer Template Only
boxheader.size Box size in bytes including the header. 1 indicates 64-bit size in size64 follow [risk: high]	ftyp.major_brand [DIFF] Orig: Enum: {isom, iso2, iso3, iso4, iso5, iso6, mp41 (+17 more)} [risk: high] FF: Enum: {avc1, iso2, isom, mmp4, mp41, mp42, mp71 (+12 more)} [risk: high]	avc1.data_reference_index Fixed: 0x01 [risk: high]
boxheader.size64 64-bit box size present when size==1; must equal actual box length including hea [risk: high]	hdlr.type [DIFF] Orig: Enum: {mhlr, dhlr, } [risk: medium] FF: Fixed: 0x68646C72 [risk: high]	avc1.depth Fixed: 0x18 [risk: medium]
boxheader.type Enum: {ftyp, moov, mvhd, trak, tkhd, edts, elst (+39 more)} [risk: high]	mdhd.version [DIFF] Orig: Enum: {0, 1} [risk: high] FF: Fixed: 0x00 [risk: high]	avc1.frame_count Fixed: 0x01 [risk: medium]
ftyp.minor_version Range: [0, 4294967295] [risk: low]	mvhd.version [DIFF]	avc1.horizresolution Fixed: 0x480000 [risk: low]

	Orig: Enum: {0, 1} [risk: high] FF: Fixed: 0x00 [risk: high]	
hdlr.subtype Enum: {vide, soun, hint, meta, subt, text, auxv (+1 more)} [risk: high]	stsd.entry_count [DIFF] Orig: Number of sample description entries in stsd box - must match number of mp4box e [risk: high] FF: Fixed: 0x01 [risk: high]	avc1.vertresolution Fixed: 0x480000 [risk: low]
mdhd.language Packed ISO-639-2/T language code (3 lowercase letters), each letter encoded as 5 [risk: low]	tkhd.version [DIFF] Orig: Enum: {0, 1} [risk: high] FF: Fixed: 0x00 [risk: high]	avcC.configurationVersion Fixed: 0x01 [risk: high]
qtgfxmode Enum: {0=qtgCopy, 32=qtgBlend, 36=qtgTransparent, 64=qtgDitherCopy, 256=qtgStraightAlpha (+4 more)} [risk: medium]		avcC.reserved_lengthSizeMinusOne Enum: {252, 253, 254, 255} [risk: medium]
stco.chunk_offset Range: [0, 4294967295] [risk: high]		avcC.reserved_numOfSequenceParameterSets Fixed: 0xE1 [risk: medium]
stsz.sample_size If non-zero, all samples have this size; if zero, individual entry_size[] table [risk: high]		esds.sectionID_3 Fixed: 0x03808080 [risk: high]
tkhd.reserved Fixed: 0x00000000 [risk: low]		esds.sectionID_4 Fixed: 0x04808080 [risk: high]
tkhd.reserved2 Fixed: 0x0000000000000000 [risk: low]		esds.sectionID_5 Fixed: 0x05808080 [risk: high]
tkhd.reserved3 Fixed: 0x0000 [risk: low]		esds.sectionID_6 Fixed: 0x06808080 [risk: high]
		ftyp.compatible_brands Enum: {isom, iso2, avc1, mp41} [risk: medium]
		ftyp.major_version Fixed: 0x0200 [risk: medium]
		ftyp.type Fixed: 0x66747970 [risk: high]
		hdlr.handler_type_audio Fixed: 0x736F756E [risk: high]
		hdlr.handler_type_video Fixed: 0x76696465 [risk: high]
		hdlr.pre_defined Fixed: 0x00 [risk: medium]
		hdlr.reserved_audio Fixed: 0x00 [risk: medium]
		hdlr.reserved_video Fixed: 0x6170706C00000000 [risk: medium]
		mdhd.type Fixed: 0x6D646864 [risk: high]
		mp4a.channelcount Fixed: 0x02 [risk: medium]
		mp4a.data_reference_index Fixed: 0x01 [risk: high]
		mp4a.samplesize Fixed: 0x10 [risk: medium]
		mvhd.type Fixed: 0x6D766864 [risk: high]
		size Box size - must be set to actual box length after content is written [risk: high]
		tkhd.duration_tkhd Range: [3, 4294967295] [risk: medium]
		tkhd.type

		Fixed: 0x746B6864 [risk: high]
		vmhd.graphics_mode Fixed: 0x00 [risk: medium]

PCAP

Original constraints: 15 FormatFuzzer constraints: 15 Only in orig: 5 In both (same): 1 In both (diff): 9 Only in FF: 5

Original Template Only	In Both	FormatFuzzer Template Only
Layer_4.tcp_hdr_len Range: [5, 15] [risk: high]	PCAPHEADER.magic_number Fixed: 0xA1B2C3D4 [risk: high]	Dot1q.L3type Fixed: 0x0800 [risk: high]
PCAPHEADER.sigfigs Fixed: 0x00000000 [risk: low]	Layer_2.L3type [DIFF] Orig: Enum: {2048=IPv4, 2054=ARP, 34525=IPv6, 33024=802.1Q} [risk: high] FF: Enum: {2048=IPv4, 33024=802.1Q VLAN} [risk: high]	Layer_4_TCP.data_offset Fixed: 0x05 [risk: high]
PCAPHEADER.thiszone Fixed: 0x00000000 [risk: low]	Layer_3.L4proto [DIFF] Orig: Enum: {1=ICMP, 6=TCP, 17=UDP, 47=GRE, 50=ESP (+3 more)} [risk: high] FF: Enum: {6=TCP, 17=UDP} [risk: high]	Layer_4_UDP.udp_hdr_len UDP datagram length - must match payload length + 8 [risk: high]
PCAPRECORD.incl_len Bytes of packet stored in file - must equal length of packet data following the [risk: high]	Layer_3.ip_hdr_len [DIFF] Orig: Range: [5, 15] [risk: high] FF: Fixed: 0x05 [risk: high]	PCAPRecord.incl_len Captured packet length - must match actual frame size [risk: high]
PCAPRECORD.orig_len Range: [0, 4294967295] [risk: medium]	Layer_3.total_length [DIFF] Orig: Total IPv4 datagram length - must be >= header length and consistent with captur [risk: high] FF: Range: [40, 104] [risk: medium]	PCAPRecord.orig_len Original packet length - should match incl_len [risk: high]
	Layer_3.version [DIFF] Orig: Enum: {4, 6} [risk: high] FF: Fixed: 0x04 [risk: high]	
	PCAPHEADER.network [DIFF] Orig: Enum: {0, 1, 101, 105, 113, 127, 147 (+2 more)} [risk: high] FF: Enum: {1=LINKTYPE_ETHERNET, 101=LINKTYPE_RAW} [risk: high]	
	PCAPHEADER.snaplen [DIFF] Orig: Range: [1, 262144] [risk: medium] FF: Range: [256, 1024] [risk: medium]	
	PCAPHEADER.version_major [DIFF] Orig: Fixed: 0x0002 [risk: high] FF: Fixed: 0x02 [risk: high]	
	PCAPHEADER.version_minor [DIFF] Orig: Fixed: 0x0004 [risk: high] FF: Enum: {2, 4} [risk: medium]	

PNG

Original constraints: 20 FormatFuzzer constraints: 19 Only in orig: 1 In both (same): 17 In both (diff): 2 Only in FF: 0

Original Template Only	In Both	FormatFuzzer Template Only
PNG_COMPR_METHOD.itxtComprMethod	PNG_CHUNK.crc CRC-32 of chunk type and data - must match calculated value	

Fixed: 0x00 [risk: medium]	[risk: high]	
	PNG_CHUNK.length Chunk data length - must match actual data byte count [risk: high]	
	PNG_CHUNK.type Enum: {IHDR, PLTE, IDAT, IEND, tRNS, cHRM, gAMA (+15 more)} [risk: high]	
	PNG_CHUNK_IHDR.bits Enum: {1, 2, 4, 8, 16} [risk: high]	
	PNG_CHUNK_IHDR.color_type Enum: {0=GrayScale, 2=TrueColor, 3=Indexed, 4=AlphaGrayScale, 6=AlphaTrueColor} [risk: high]	
	PNG_CHUNK_IHDR.compr_method Fixed: 0x00 [risk: high]	
	PNG_CHUNK_IHDR.filter_method Fixed: 0x00 [risk: high]	
	PNG_CHUNK_IHDR.interlace_method Enum: {0=NoInterlace, 1=Adam7Interlace} [risk: high]	
	PNG_CHUNK_TIME.timeDay Range: [1, 31] [risk: low]	
	PNG_CHUNK_TIME.timeHour Range: [0, 23] [risk: low]	
	PNG_CHUNK_TIME.timeMin Range: [0, 59] [risk: low]	
	PNG_CHUNK_TIME.timeMonth Range: [1, 12] [risk: low]	
	PNG_CHUNK_TIME.timeSec Range: [0, 60] [risk: low]	
	PNG_SIGNATURE.btPngSignature[0] Fixed: 0x8950 [risk: high]	
	PNG_SIGNATURE.btPngSignature[1] Fixed: 0x4E47 [risk: high]	
	PNG_SIGNATURE.btPngSignature[2] Fixed: 0x0D0A [risk: high]	
	PNG_SIGNATURE.btPngSignature[3] Fixed: 0x1A0A [risk: high]	
	PNG_CHUNK_IHDR.height [DIFF] Orig: Range: [1, 2147483647] [risk: high] FF: Range: [1, 24] [risk: medium]	
	PNG_CHUNK_IHDR.width [DIFF] Orig: Range: [1, 2147483647] [risk: high] FF: Range: [1, 24] [risk: medium]	

WAV

Original constraints: 17 FormatFuzzer constraints: 17 Only in orig: 2 In both (same): 4 In both (diff): 11 Only in FF: 2

Original Template Only	In Both	FormatFuzzer Template Only
FORMATCHUNK.dwAvgBytesPerSec Average bytes per second = dwSamplesPerSec * wBlockAlign [risk: high]	DATA_CHUNK.chunkID Fixed: 0x64617461 [risk: high]	CUE_CHUNK.dwCuePoints Range: [1, 16] [risk: medium]
FORMATCHUNK.dwSamplesPerSec Enum: {8000, 11025, 16000, 22050, 32000, 44100, 48000 (+4 more)} [risk: high]	FORMATCHUNK.chunkID Fixed: 0x666D7420 [risk: high]	FACT_CHUNK.chunkSize Fixed: 0x0C [risk: high]
	WAVRIFFHEADER.groupID Fixed: 0x52494646 [risk: high]	

	WAVRIFFHEADER.riffType Fixed: 0x57415645 [risk: high]	
	CUECHUNK.chunkID [DIFF] Orig: Fixed: 0x63756520 [risk: medium] FF: Fixed: 0x63756520 [risk: high]	
	DATAHUNK.chunkSize [DIFF] Orig: Size of waveform sample data - must be a multiple of wBlockAlign [risk: high] FF: Data chunk size - must match length of PCM payload [risk: high]	
	FACTHUNK.chunkID [DIFF] Orig: Fixed: 0x66616374 [risk: medium] FF: Fixed: 0x66616374 [risk: high]	
	FORMATHUNK.chunkSize [DIFF] Orig: Enum: {16, 18, 20, 40} [risk: high] FF: Fixed: 0x10 [risk: high]	
	FORMATHUNK.wBitsPerSample [DIFF] Orig: Enum: {8, 16, 24, 32} [risk: high] FF: Enum: {8, 16, 32} [risk: high]	
	FORMATHUNK.wBlockAlign [DIFF] Orig: Block alignment = wChannels * wBitsPerSample / 8 [risk: high] FF: Block alignment - must equal (wBitsPerSample / 8) * wChannels [risk: high]	
	FORMATHUNK.wChannels [DIFF] Orig: Range: [1, 8] [risk: high] FF: Range: [1, 256] [risk: medium]	
	FORMATHUNK.wFormatTag [DIFF] Orig: Enum: {1=PCM, 2=ADPCM, 3=IEEE_FLOAT, 6=ALAW, 7=MULAW (+2 more)} [risk: high] FF: Enum: {1=PCM} [risk: high]	
	LISTHUNK.chunkID [DIFF] Orig: Fixed: 0x4C495354 [risk: medium] FF: Fixed: 0x4C495354 [risk: high]	
	SMPLCHUNK.chunkID [DIFF] Orig: Fixed: 0x736D706C [risk: medium] FF: Fixed: 0x736D706C [risk: high]	
	WAVRIFFHEADER.hsize [DIFF] Orig: Total RIFF chunk size in bytes - should equal file size minus 8 [risk: medium] FF: RIFF chunk size - total file size minus 8 [risk: high]	

ZIP

Original constraints: 22 FormatFuzzer constraints: 21 Only in orig: 12 In both (same): 7 In both (diff): 3 Only in FF: 11

Original Template Only	In Both	FormatFuzzer Template Only
AESMODE Enum: {1, 2, 3} [risk: high]	ZIP64ENDLOCATOR.elSignature Fixed: 0x07064B50 [risk: high]	ZIPDIRENTRY.deCompressedSize Must mirror the matching local file record compressed size [risk: high]
ALGFLAG	ZIP64ENDLOCATORRECORD.elr64Signature	ZIPDIRENTRY.deCompression

Enum: {26113, 26114, 26115, 26121, 26126, 26127, 26128 (+5 more)} [risk: high]	Fixed: 0x06064B50 [risk: high]	Must mirror the matching local file record compression method [risk: high]
COMPTYPE Enum: {0, 1, 2, 3, 4, 5, 6 (+12 more)} [risk: high]	ZIPDATADESCR.ddSignature Fixed: 0x08074B50 [risk: high]	ZIPDIRENTRY.deCrc Must mirror the matching local file record CRC [risk: high]
EXTRAFIELD.efDataSize Size of extra field data section in bytes - must match length of efData payload [risk: high]	ZIPDIGITALSIG.dsSignature Fixed: 0x05054B50 [risk: high]	ZIPDIRENTRY.deDiskNumberStart Fixed: 0x00 [risk: medium]
EXTRAFIELD.efHeaderID Enum: {1, 7, 8, 9, 10, 12, 13 (+39 more)} [risk: medium]	ZIPDIRENTRY.deSignature Fixed: 0x02014B50 [risk: high]	ZIPDIRENTRY.deHeaderOffset Offset of matching local file header from start of archive [risk: high]
HOSTOSTYPE Enum: {0=FAT, 1=AMIGA, 2=VMS, 3=Unix, 4=VM_CMS (+15 more)} [risk: medium]	ZIPENDLOCATOR.elSignature Fixed: 0x06054B50 [risk: high]	ZIPDIRENTRY.deUncompressedSize Must mirror the matching local file record uncompressed size [risk: high]
PRCFLAG Enum: {1, 2, 3} [risk: medium]	ZIPFILERECD.frSignature Fixed: 0x04034B50 [risk: high]	ZIPENDLOCATOR.elDiskNumber Fixed: 0x00 [risk: medium]
SignatureTYPE Enum: {67324752=S_ZIPFILERECD (0x04034B50 'PK\x03\x04'), 134695760=S_ZIPDATADESCR (0x08074B50 'PK\x07\x08'), 33639248=S_ZIPDIRENTRY (0x02014B50 'PK\x01\x02'), 84233040=S_ZIPDIGITALSIG (0x05054B50 'PK\x05\x05'), 101075792=S_ZIP64ENDLOCATORRECORD (0x06064B50 'PK\x06\x06') (+2 more)} [risk: high]	ZIP64ENDLOCATORRECORD.elr64DirectoryRecordSize [DIFF] Orig: Size of the ZIP64 end of central directory record minus 12 (the size + signature [risk: high]) FF: Range: [0, 52] [risk: medium]	ZIPENDLOCATOR.elStartDiskNumber Fixed: 0x00 [risk: medium]
WzAES.VendorID Fixed: 0x4145 [risk: high]	ZIPFILERECD.frCompressedSize [DIFF] Orig: Length in bytes of compressed file data following the header - must match actual [risk: high] FF: Compressed payload size [risk: high]	ZIPFILERECD.frCompression Enum: {0=COMP_STORED, 8=COMP_DEFLATE} [risk: high]
ZIPENDLOCATOR.elCommentLength Length of the .ZIP file comment - must equal actual size of elComment field [risk: medium]	ZIPFILERECD.frFileNameLength [DIFF] Orig: Length of file name - must equal actual size of frFileName field [risk: high] FF: Length of the file name in bytes [risk: high]	ZIPFILERECD.frCrc CRC-32 of uncompressed data - must match payload [risk: high]
ZIPENDLOCATOR.elDirectoryOffset Offset of start of central directory - must point to a valid ZIPDIRENTRY signatu [risk: high]		ZIPFILERECD.frUncompressedSize Uncompressed payload size [risk: high]
ZIPFILERECD.frExtraFieldLength Sum of EXTRAFIELD (efDataSize + 4) entries must equal frExtraFieldLength [risk: high]		